Research Design Memo:
EU AI Regulation Timing and U.S. Federal AI/Cloud Procurement

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Attached materials.

- **Code:** `code/analysis_final_memo_authentic.py` — ingestion, cleaning, panel build, OLS with firm fixed effects and HC1 standard errors.
- **Prompts:** `prompts/final_memo_prompts.md` — optional LLM prompt for GRI v2, memo-alignment notes, data-handling note.

Executive Summary

This memo examines whether the timing of major EU AI Act milestones lines up with changes in realized U.S. federal procurement flows to selected AI and cloud vendors. The analysis combines three ingredients: a quarter-level regulatory timing indicator tied to the European Parliament's June 2023 vote, transaction-level USASpending extracts covering prime and sub-award activity, and a simple text-based "geopolitical rhetoric index" (GRI) built from contract descriptions. After aggregating 2,579 transactions into a 70-cell firm–quarter panel spanning 2019Q1–2025Q3, the analysis shows a clear upward shift in average quarterly obligations after 2023Q2, especially for AWS, and an OLS specification with firm fixed effects and HC1 standard errors returns a positive, statistically significant coefficient on the post-shock indicator. The lagged keyword-based GRI, by contrast, is imprecisely estimated and adds little explanatory power. I interpret these results as descriptive evidence about timing and portfolio composition rather than a clean causal estimate, because the same period also captures large concurrent shifts in U.S. defense cloud spending and procurement modernization.

1 The Political Risk Question

Core question:

> How does **transatlantic regulatory divergence** around AI—specifically the **timing of EU AI Act legislative milestones**—relate to **realized U.S. federal procurement flows** (prime and sub-award obligations) to **selected AI and cloud vendors**, conditional on observable **mission-oriented framing** in contract narratives?

The political signal in the design is a binary EU "shock" calendar anchored to the European Parliament's **June 2023 negotiating-position vote**, after which trilogue and adoption dynamics moved much more quickly toward a final AI Act framework. The economic outcome is quarterly aggregate federal obligations to named recipients in USASpending-style transaction extracts, measured in U.S. dollars at the firm–calendar-quarter level.

This framing keeps the question close to market-relevant quantities. If the "Brussels Effect" pushes vendors toward common compliance standards, product redesign, or geographic segmentation, investors would want to know whether those pressures coincide with weaker public-sector demand, offsetting demand, or simply a reallocation across business lines. U.S. national-security procurement is therefore useful not only as an outcome in its own right, but also as a possible hedge channel against regulatory pressure coming from Europe.

2 Existing Approaches to Answering This Question

Several literatures and industry practices bear on the link between **foreign AI governance** and **firm-level revenue exposure**, though few combine EU milestones with granular U.S. obligating actions.

2.1 "Brussels Effect" and extraterritorial regulation. Legal and IR scholarship (e.g., Bradford's framing of EU regulatory export) emphasizes how EU rules reshape global product standards when firms unify compliance to access the Single Market. Empirical work often studies firm adoption of privacy or product rules or equity-market reactions to headline regulatory events. For AI, analysts track draft text, trilogue outcomes, and final risk-tier lists; event-study abnormal returns around key votes are common in sell-side and policy notes.

2.2 Event studies and difference-in-differences. Applied work in law, economics, and CS policy uses staggered adoption, EU vs. non-EU comparisons, or sector exposure scores from 10-K and supply-chain data. Outcomes include stock returns, patenting, job postings, or Model lists compliance. Some designs exploit national transposition timing or court decisions.

2.3 Vendor-level public procurement analytics. Researchers use USASpending, FPDS, or SAM data to trace award families, IDIQs, and agency programs (e.g., cloud GWACs). Practitioners build vendor revenue bridges from transaction-level obligations to investor filings where possible. Text fields are mined for NAICS, PSC, program acronyms, and mission language.

2.4 Text-as-data measures. Political science and computational social science extract frames from speech, media, or corporate disclosure. A simple keyword index is a baseline; dictionaries, topic models, and increasingly LLM structured extraction reduce polysemy but raise reproducibility and PII concerns with raw government text.

2.5 Sell-side "policy beta" and ESG overlays. Equity research teams sometimes construct explicit exposure scores to data localization, AI ethics, or sanctions, then correlate them with returns or forward estimates. These overlays are valuable for portfolio-facing summaries but often aggregate at sector level or rely on subjective disclosure coding unless paired with transaction facts such as obligations.

Taken together, these literatures point toward a hybrid design: use EU legislative timing to define the political backdrop, use procurement transactions to observe realized commercial exposure, and use text analysis to recover variation in how awards are framed. None of those pieces is sufficient on its own, but combined they make it possible to study a vendor-level channel that is usually discussed only in broad sector terms.

3 Limits of Existing Approaches (for This Question)

3.1 Identification. EU AI Act milestones are not exogenous shocks to U.S. procurement; the same period saw JWCC rollout, Ukraine conflict–driven modernization, CHIPS-era industry policy, and Fed tightening. A calendar dummy after a Parliament vote picks up joint trends, not marginal effects of Brussels on U.S. obligation streams.

3.2 Cross-border mapping. The Brussels Effect predicts global product harmonization, not necessarily lower U.S. public spending. Firms may dual-track offerings: EU-compliant SKUs vs. U.S.-only missions. Aggregate equity reactions may miss segment-level procurement.

3.3 Data granularity vs. noise. USASpending transaction obligations vary with modification timing, IDIQ plumbing, and reporting errors. Single-line outliers can dominate small vendor panels. Sub-award descriptions may be thin or mis-keyed.

3.4 Text measurement. Keyword "security" hits cybersecurity patches as often as national security. Without context-aware scoring, GRI v1 is high-noise.

3.5 Coverage. Focusing on three firms yields a proof-of-concept, not representative inference for AI-adjacent federal suppliers.

3.6 Temporal misalignment. Contract action dates reflect ordering activity, not necessarily EU board decisions or product redesign completion. A vendor may obligate under an IDIQ option years after an EU vote; conversely, regulatory anticipation may shift behavior before the milestone encoded here.

3.7 Firm FE vs. firm-specific trends. With T quarters modest and N firms tiny, linear trends per firm may soak variance now attributed to D_t; conversely, firm FE alone do not absorb common macro shocks correlated with the EU timeline.

4 Prototype Research Design: "PR–Procurement Tracker"

In response to those limits, the prototype focuses on transparency and tractability rather than elaborate identification. The workflow is intentionally simple enough to audit line by line while still being detailed enough to show how a richer procurement-risk tracker could be built. Concretely, it:

1. Reads firm-specific USASpending CSV extracts under `data` covering prime contracts, prime assistance, and sub-awards, then harmonizes the relevant identifier, date, amount, and description fields across file types.

2. Restricts the sample to four validated UEIs that map to three firms: one UEI for Palantir, two UEIs merged into C3.ai, and one UEI for AWS.

3. Converts action dates to calendar quarters, drops records with missing quarter information, and trims any single-line obligation above 10^{10} USD as an obvious reporting artifact.

4. Constructs GRI v1 as the share of transactions in each firm–quarter whose description contains at least one keyword from a fixed list (*defense, security, intelligence, warfighting, surveillance*).

5. Lags GRI by one quarter within firm so that the text signal is temporally prior to the procurement outcome it is used to explain.

6. Defines $D_t = 1$ for 2023Q3 onward and $D_t = 0$ beforehand, matching the June 2023 parliamentary milestone at the quarter level.

7. Estimates a firm–quarter OLS regression with firm fixed effects implemented as indicator variables; under `pandas drop_first` ordering, AWS is the omitted reference category.

Main estimating equation. Let i index **firm** and t index **calendar quarter**. The dependent variable Y_{it} is obligations in **millions USD** in quarter t; α_i denotes firm fixed effects (one firm omitted); $\mathrm{GRI}_{i,t-1}$ is the one-quarter lag of the geopolitical rhetoric index within firm i; D_t is the EU milestone / post-shock dummy (common across firms in quarter t); and ε_{it} is the idiosyncratic error. The specification is

$$Y_{it} = \beta_0 + \alpha_i + \beta_1 \, \mathrm{GRI}_{i,t-1} + \beta_2 D_t + \varepsilon_{it} \,. \tag{1}$$

Definition of $\mathrm{GRI}_{i,t}$. Let m_{it} be the number of transactions in firm i, quarter t, with at least one keyword match, and let n_{it} be the total number of transactions in (i, t). Then

$$\mathrm{GRI}_{i,t} = \frac{m_{it}}{n_{it}} \,. \tag{2}$$

The regressor $\mathrm{GRI}_{i,t-1}$ is this share lagged one calendar quarter within each firm.

This design is best understood as a structured first pass: it standardizes heterogeneous procurement files, produces a panel that can be inspected directly, and makes the modeling assumptions unusually transparent. It is still not a final causal estimator, but it offers a concrete empirical baseline instead of a purely qualitative discussion of regulatory risk.

Operational workflow. The script `code/analysis_final_memo_authentic.py` recursively ingests CSV files, dispatches each file to a loader based on whether it is a prime-contract, prime-assistance, or sub-award extract, standardizes column names to a common schema, coerces obligations to numeric values, fills missing descriptions with empty strings, filters to the validated UEI list, and drops rows without a parsable action quarter. It then aggregates transactions to the firm–quarter level, counts both total transactions and keyword matches, computes the contemporaneous GRI share and its within-firm lag, rescales obligations into millions of dollars, and estimates OLS with HC1 heteroskedasticity-robust covariance. Because the panel is small and visibly unbalanced across firms, the script also prints pre/post means so that the regression output can be interpreted against the raw descriptive shift rather than in isolation.

Contribution relative to Sec. 2. The contribution here is not a new theory of the Brussels Effect. It is a transparent bridge between an external regulatory timeline and observable U.S. procurement outcomes at vendor granularity, plus a rudimentary text measure that can later be replaced with a more context-aware GRI v2.

5 Preliminary Data Analysis and Proof-of-Concept Results

Sample and span. The cleaned transaction file contains 2,579 observations from 2019Q1 through 2025Q3. After aggregation, those records produce 70 observed firm–quarter cells: 27 for AWS, 27 for Palantir, and 16 for C3.ai. Vendor identities are defined by validated UEIs: Palantir (`FSY4LVSBGWB7`), C3.ai (`YT9GYGR24NW9` and `YYRJSL45NMQ7` combined), and AWS (`NQEWN6C1LSU5`). The panel is therefore informative about within-firm time variation, but noticeably thinner for C3.ai than for the other two firms.

Firm	Pre-shock mean ($M/qtr)	Post-shock mean ($M/qtr)
AWS	∼22.7	∼125.3
Palantir	∼45.6	∼68.5
C3.ai	∼1.5	∼3.3

Descriptive shift (mean quarterly obligations, USD millions). The first pattern in the data is a strong pre/post level shift around 2023Q3. After removing the single implausible Palantir sub-award artifact, mean quarterly obligations rise for all three firms:
These means are calculated over the quarters observed for each firm. Because the panel is unbalanced, they should be read as descriptive summaries of realized activity rather than as directly comparable exposure rates. In level terms, AWS accounts for the largest post-shock expansion: total obligations rise from roughly $0.41B before 2023Q3 to about $1.13B afterward. Palantir remains large in both periods, with about $0.82B pre-shock and $0.62B post-shock, but its post-shock mean still increases because the later period covers fewer quarters with larger average actions. C3.ai remains much smaller in absolute dollars, moving from roughly $12M pre-shock to $26M post-shock.

Qualitative anchors (major vehicles).

- **AWS:** Joint Warfighting Cloud Capability (JWCC) language on DISA/WHS vehicles (e.g., family `HC105023D0005`).
- **Palantir:** Army Capability Drop 1 delivery orders under Army `W56KGY18D0003` (2025 obligations).
- **C3.ai:** Air Force SBIR Phase III database support (`FA460019CA025`) with modifications referencing SBIR Phase III.

These qualitative anchors help explain why the post-2023Q2 increase is not naturally attributable to EU regulation alone. Much of the visible growth in the procurement data is tied to mission programs, cloud vehicles, and defense modernization pathways that have their own U.S. budget and operational logic.

Regression (HC1 robust SE). The estimating sample falls from 70 to $N = 67$ firm–quarters once the one-quarter lag of GRI is imposed. AWS is the reference firm because its indicator is omitted from the regression design matrix.

Variable	Coef.	Robust SE	z	p
Constant	42.319	8.984	4.71	0.000
$\text{GRI}_{i,t-1}$	11.838	26.040	0.46	0.649
D_t	**47.187**	13.709	**3.44**	**0.001**
C3.ai vs. AWS	−66.611	15.254	−4.37	0.000
Palantir vs. AWS	−5.667	13.806	−0.41	0.681

$R^2 = 0.358$; Adj. $R^2 = 0.317$.

Interpretation. Several features of the regression are worth separating. First, the positive estimate on D_t (about 47.2) means that, conditional on firm fixed effects and lagged GRI, post-2023Q2 quarters are associated with roughly $47M higher obligations per firm–quarter. That coefficient is statistically precise in this small sample, but the precision should not be mistaken for identification: the same post period captures expanding U.S. defense cloud procurement, especially in AWS-linked vehicles, and broader procurement normalization after earlier disruptions. Second, the lagged GRI coefficient is positive in sign but far from statistically distinguishable from zero. In practical terms, the keyword index does not explain much additional quarter-to-quarter variation once firm baselines and the common post period are already in the model. That is consistent with the structure of the raw data: keyword hit rates are relatively low for AWS (about 3.6% of transactions overall), modest for Palantir (about 6.9%), and higher for C3.ai (20%) but on a very small transaction base. Third, the firm coefficients mainly recover level differences in the panel rather than deep structural contrasts. C3.ai sits far below AWS in obligation levels, while Palantir is statistically similar to AWS after the common post dummy is included. Overall, the regression is most useful as a compact summary of timing patterns already visible in the descriptive data, not as a decisive test of a Brussels-driven mechanism.

6 Discussion: Brussels Effect vs. U.S. Procurement Channel

EU channel: Stricter rules on biometric surveillance and high-risk AI can raise compliance costs and product segmentation for vendors operating in Europe.
U.S. channel: Obligation narratives emphasize warfighting cloud, Army intelligence modernization, and SBIR Phase III pathways—independent of Brussels timelines.
Joint takeaway: In this sample, regulatory divergence appears more likely to reshape product strategy, compliance cost, and geographic segmentation than to mechanically suppress U.S. procurement flows. The evidence points to overlapping channels rather than a single dominant effect, which is why a scenario framework is more credible here than a single reduced-form causal claim.

7 Next Steps

7.1 Causal front door: Merge firm-level EU revenue shares with fine-grained AI Act instruments; estimate triple-differences (high-EU exposure $\times$ high-surveillance product $\times$ post-milestone) with firm and calendar FE, or Sun–Abraham staggered designs where appropriate.

7.2 Parallel trends diagnostics: Plot firm-specific cumulative obligations pre 2023Q3 against placebo shock dates; run TWFE bias diagnostics when D_t is essentially common time.

7.3 Expand the cross-section: Additional cloud/AI primes with consistent UEI parenting rules (`recipient_parent_uei` vs. immediate recipient).

7.4 GRI v2 with LLMs: Freeze model ID, temperature, and prompt hash; label unique descriptions once; human audit on a stratified sample (see `prompts/final_memo_prompts.md`).

7.5 Outliers and linkage: Award-family ceilings from parent PIID distributions; join to FPDS for pricing and award-type filters.

7.6 Forecasting track: Lag-GRI, agency dummies, and macro controls to forecast next-quarter obligations out-of-sample; report RMSE by firm.

8 Compliance and Reproducibility

All numerical results in Sec. 5 were produced by running `python code/analysis_final_memo_authentic.py` from the project root on the `data` extracts bundled with this project. The current pipeline does not apply an LLM to raw federal text: text processing is limited to deterministic keyword matching with a fixed regex compiled from the GRI dictionary. Optional prompts for a future GRI v2 workflow are stored in `prompts/final_memo_prompts.md`.

End of research design memo.